

Performance Testing

Date	02 October 2026
Team ID	SWTID-2026-2606
Project Name	PocketSmart AI
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how PocketSmart AI behaves under expected and peak load conditions. This document records the planned performance-testing approach for the FastAPI backend and identifies the measurements to capture during execution.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter (performance/load testing)
Type of Testing	Load Testing
Target Module / API	FastAPI APIs: /api/generate-home and /api/shopping
Test Environment	Local development environment (FastAPI + Uvicorn)
Test Date	02 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home Planner recommendation request under normal load	5	30	Successful responses with acceptable response time and no server errors.
2	Shopping/Product Discovery request with category, budget and platform filters	10	60	Filtered products are returned correctly without request failures.
3	Mixed planner/API requests under increased load	20	60	Application remains responsive and error rate stays below the target.
4	Peak-load API test across recommendation endpoints	50	60	Server handles peak requests without crashes or significant degradation.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Not measured	Not Tested	Dedicated performance run not yet executed.
2	Response Time (Max)	< 5 seconds	Not measured	Not Tested	Dedicated performance run not yet executed.
3	Throughput (Req/sec)		Not measured	Not Tested	Throughput requires an executed load test.
4	Error Rate	< 1%	Not measured	Not Tested	No performance-test error-rate data is available yet.
5	CPU Utilization	< 80%	Not measured	Not Tested	CPU utilization must be captured during the load test.
6	Memory Utilization	< 80%	Not measured	Not Tested	Memory utilization must be captured during the load test.

Step 4: Observations & Analysis

Current status: The application has been functionally tested locally, including the FastAPI endpoints and shopping filters. A dedicated load/performance run has not yet been executed, so response time, throughput, CPU, memory, and error-rate values are marked as Not Tested rather than estimated.

Key Findings:

[Describe what the test results revealed about your system's performance]

Bottlenecks Identified:

[List any performance bottlenecks observed during testing]

Optimization Steps Taken:

[Describe any changes or optimizations made to improve performance]